

Beyond Implied Volatility: Option-Implied Entropy and the Predictability of Risk

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Abstract

Option prices tell us more than how volatile the market expects the S&P 500 to be: they reveal the shape of the whole range of outcomes investors are pricing. This project asks whether that shape helps forecast future volatility once we already know the market's volatility forecast, implied volatility. I extract the market-implied distribution of S&P 500 returns from option prices every trading day from 2015 to 2025, and measure its shape with excess entropy: how much its spread differs from a standard lognormal distribution of the same width. I then test whether adding this measure improves forecasts of volatility over the following 21 trading days, using only information available at the time of each forecast (January 2018 to July 2025, 1,905 days). It does not. Adding entropy slightly worsens the forecasts (out-of-sample R^2 falls by 0.011; Clark–West $t = -1.76$), whereas implied volatility improves them substantially (R^2 rises by 0.139; $t = +4.77$). Two other shape measures, risk-neutral skewness and kurtosis, add nothing either. A simulation shows the test would detect an effect worth about 0.018 of R^2 80% of the time, so a meaningful effect is unlikely to have been missed. The result holds at forecast horizons from 5 to 63 days, without 2020, and in a separately written version of the test. A fresh review before submission found three further errors (look-ahead in the training data, zero-bid option quotes distorting the tails, and mislabelled statistics); the conclusion survived each correction.

1 Introduction

Implied volatility, the volatility priced into options, is one of the best available forecasts of how much the stock market will move. But it only describes the width of the range of outcomes the market expects. Option prices across many strike prices reveal the full shape of that range: how heavy the tails are, and how lopsided it is towards crashes. If investors price in risks before those risks show up in actual market moves, the shape might carry information that width alone misses.

I test this with one measure of shape, excess entropy. Entropy measures how spread out a distribution is. Comparing the entropy of the market-implied distribution with that of a lognormal distribution of the same width, the shape assumed by the Black–Scholes model (Black and Scholes 1973), isolates the part of the shape that a standard model would not produce.

The test is set up to give a clear answer:

- **The question.** Does excess entropy improve forecasts of S&P 500 volatility over the next 21 trading days, beyond past volatility (the HAR model), implied volatility, and two standard shape measures (BKM skewness and kurtosis)?

- **A fair comparison.** Before entropy is tested, the part of it that simply repeats implied volatility and the other measures is removed (orthogonalisation), so it is judged only on what it adds.
- **Checks on the test itself.** Before any entropy result was accepted, the pipeline had to reproduce a known finding: that implied volatility forecasts better than past volatility. A placebo made of pure noise, and a simulation of the test’s power, show what the test can and cannot detect.

The answer is no. Adding entropy slightly worsens out-of-sample forecasts ($R^2 - 0.011$, $t = -1.76$), and the simulation shows that an effect worth about 0.018 of R^2 would be detected 80% of the time. Skewness and kurtosis add nothing either. In this sample, once you know how wide the market expects the range of outcomes to be, its shape tells you nothing more about future volatility.

The project also shows why such checks matter. The built-in check caught five data and code errors during the summer, and a fresh review before submission caught three more. Section 8 lists all eight, because each would have produced a plausible but wrong answer.

2 Related literature

Four strands of work frame the test.

- **Implied versus realised volatility.** Christensen and Prabhala (1998) show that implied volatility forecasts realised volatility and largely subsumes past realised volatility. This result is the calibration gate used here: a pipeline that cannot reproduce it is broken.
- **Realised volatility forecasting.** Corsi (2009) introduces the HAR-RV model, which captures the long memory of volatility with daily, weekly and monthly lags. It is the standard benchmark any new predictor must beat. Patton (2011) shows which loss functions (MSE on variance, QLIKE) give robust rankings when the target is a noisy proxy.
- **Model-free implied moments.** Bakshi, Kapadia and Madan (2003) derive risk-neutral skewness and kurtosis directly from OTM option prices. These are the natural competitors for any shape measure, since entropy and higher moments overlap heavily.
- **Entropy and the RND.** Buchen and Kelly (1996) recover maximum-entropy densities from option prices; Rompolis and Tzavalis (2008) and Rompolis (2010) extend density recovery and test it empirically. These papers use entropy as an extraction tool. This paper uses it instead as the state variable whose forecasting content is tested.

Gao and Martin (2021) use option prices to measure market conditions, including bubble-like episodes, that valuation ratios alone miss. The economic motivation for a shape effect runs through investor preferences over skewness: if investors pay for protection against left-tail outcomes, the RND departs from lognormal, and the size of that departure could signal changing risk.

3 Data

The merged daily panel has 2,637 observations from 2015 to 2025, built from SPX options and SPY returns.

Options. End-of-day SPX option quotes come from OptionMetrics IvyDB via WRDS (over 18 million rows for 1996–2025; the 2015–2025 sample uses 6,757 date-expiry pairs of AM-settled

monthly options). The forward for each expiry is implied from put–call parity at the strike where call and put mid-prices are closest, and the BKM moments are computed with a zero interest rate, a simplification worth about 0.4% of option value at 30 days and 5% rates. Only strikes with $0.4 < K/F < 1.6$ and a mid-price above \$0.05 are used. Filters, fixed before estimation:

- Maturities of 14 to 90 calendar days; each date keeps the single expiry closest to 30 days, matching the 21-trading-day forecast horizon.
- Out-of-the-money quotes only: puts below the forward, calls above. ITM time value is swamped by the bid-ask spread.
- At least 12 usable strikes per date.

The sample starts in 2015 because SPX strike density before then is too sparse for a stable second derivative of the call curve; the strike-coverage diagnostics are in the repository (strike_density_diagnostics.pdf). The out-of-sample period is 2018 to 2025.

Realised variance. The dependent variable is the log of average daily realised variance over the next 21 trading days, excluding day t . Daily variance is the squared close-to-close return of SPY from the official session close (WRDS Intraday Indicators). Range-based estimators were tested and rejected because the high and low fields contain erroneous prints (Section 8). As a validation, $\text{corr}(\log \text{VIX}, \log \text{forward 21-day RV}) = 0.672$, inside the 0.65 to 0.72 range expected for a forward-looking target.

Zero returns. SPY closed exactly unchanged on 8 days, so $rv = 0$ and its log is undefined. Each of these days has a normal intraday range and an ordinary next-day return, so they are genuine unchanged closes rather than stale prices. Daily rv is floored at the 0.5th percentile of its positive values (2.4×10^{-9}) before logs are taken. Under the original floor of 10^{-12} , these days entered the HAR lags at $\log rv = -27.6$ against a median of -10.6 . Daily returns of 20% or more in absolute value are treated as data errors and dropped; SPY’s largest genuine move in the sample is about 11%.

Table 1: Summary statistics, daily panel, $n = 2,637$ (5 February 2015 to 31 July 2025)

Variable	Mean	SD	1st pct	Median	99th pct	AR(1)
Log forward 21-day RV (target)	−9.602	1.012	−11.621	−9.700	−6.709	0.99
Log daily RV	−11.0	2.5	−19.1	−10.6	−6.5	0.15
ATM implied volatility (% , annualised)	15.3	6.7	6.9	13.5	37.4	0.96
Excess entropy XH (nats)	−0.130	0.048	−0.252	−0.126	−0.036	0.88
Orthogonalised $XH^\perp$ (nats, full sample)	0.000	0.030	−0.090	0.005	0.056	0.77
BKM skewness	−3.63	1.31	−7.15	−3.51	−1.27	0.95
BKM kurtosis	39.7	25.4	7.7	34.1	124.2	0.94
Half-spread (% of forward)	0.013	0.013	0.003	0.008	0.058	0.80
OTM strikes used	278	88	105	280	460	0.98
Days to expiry	29.7	8.9	14	30	46	0.74

Entropy and BKM statistics are after the zero-bid filter. The target’s AR(1) of 0.99 reflects overlapping 21-day windows, not persistence in daily volatility. XH is negative on every panel date and correlates +0.43 with log IV. After orthogonalisation $XH^\perp$ is uncorrelated with log IV by construction and keeps 38% of XH ’s variance; its AR(1) here (0.77) uses the full-sample residual, whereas the power analysis uses the real-time residual from March 2016 (0.82). The BKM moments are large for a 30-day horizon but consistent with the option prices: skewness tracks the Cboe SKEW index (Section 8), and since kurtosis must exceed skewness squared plus

one, a median skewness of -3.5 implies a kurtosis of at least 13, against a median of 34. The half-spread is the median over quotes with $0.95 < K/F < 1.05$.

4 Measuring excess entropy

Excess entropy is the entropy of the extracted density minus that of a lognormal with the same mean and variance; it isolates shape from width.

Strike selection. Out-of-the-money quotes are selected with the rule Cboe uses for the VIX index (Cboe 2026): quotes with a zero bid are dropped, and each wing stops after two consecutive zero bids, moving outward from the forward. Without it, a deep put quoted 0 bid / 0.10 ask enters at a mid-price of 0.05 and overprices the tail. Section 8 shows the effect: on affected dates excess entropy fell to -0.95 nats and BKM kurtosis exceeded 500.

Density extraction. On each date, OTM implied volatilities are fitted with a cubic smoothing spline in log moneyness. The smoothed smile is re-priced into a call curve $C(K)$ on a 4,000-point grid of K/F from 0.05 to 3.0, and the density follows from Breeden and Litzenberger (1978):

$$q(K) = e^{rT} \frac{\partial^2 C(K)}{\partial K^2}. \quad (1)$$

Beyond the outermost quoted strikes, the tails are lognormal at the boundary implied volatility. The probability mass from these extrapolated tails is recorded but does not enter the regressions.

Entropy. With $x = K/F$, differential entropy (Shannon 1948) and its lognormal benchmark are

$$h(q) = - \int q(x) \ln q(x) dx, \quad h_{LN} = m + \frac{1}{2} \ln(2\pi e s^2), \quad (2)$$

$$s^2 = \ln(1 + \sigma_x^2/\mu_x^2), \quad m = \ln \mu_x - \frac{1}{2}s^2, \quad XH = h(q) - h_{LN}, \quad (3)$$

where μ_x and σ_x^2 are the mean and variance of q . A negative XH means the density is more concentrated than a lognormal of the same width, as when mass shifts into a heavy left tail and a sharper centre.

Sign of XH . XH is not bounded above by zero by construction. The lognormal is the maximum-entropy distribution for a fixed mean and variance of $\log x$, not of x , and moments here are matched in $x = K/F$. In this sample the bound never binds: XH is negative on all 2,681 dates with an extracted density (one expiry per date), with a maximum of -0.0075 nats.

Measurement error. A synthetic harness, built before touching real data, tests the extractor on densities with known closed-form entropy at realistic strike counts and quote noise. Three findings, recorded in the research log rather than reproduced by the repository, fixed the design:

1. Smoothing $s = 10^{-2}$ is the smallest value at which entropy bias and variance are stable at measured quote noise (bias of roughly $+0.01$ to $+0.06$ nats). It was locked before any regression was run.
2. Entropy is a nonlinear functional of q , so by Jensen's inequality quote noise biases it rather than averaging out, and the bias grows with the bid-ask spread, so the half-spread enters the orthogonalisation as a control.

3. Bimodality is not detectable at SPX quote precision (0% detection at every realistic noise level). Event-driven two-peak densities are outside what this method can see.

On the pre-filter extraction, the harness’s noise model predicted the real-data coefficient on log half-spread at 0.0237 against 0.0227 realised, a gap of about 4% (research log).

BKM moments. Risk-neutral skewness and kurtosis are computed model-free from the same OTM quotes following Bakshi, Kapadia and Madan (2003).

5 Forecasting design

Four nested models are compared out of sample, and entropy is only allowed to compete after its overlap with everything else has been removed.

Models. With y_t the log of average realised variance over days $t + 1$ to $t + 21$, the largest model is

$$y_t = \beta_0 + \beta_d RV_t^{(1)} + \beta_w RV_t^{(5)} + \beta_m RV_t^{(22)} + \gamma \ln IV_t + \delta_1 SKEW_t + \delta_2 KURT_t + \theta XH_t^\perp + \varepsilon_t, \quad (4)$$

where $RV_t^{(k)}$ is the average log daily RV over the last k days (Corsi 2009). The nested sequence adds one block at a time: HAR, then +IV, then +BKM skew and kurtosis, then $+XH^\perp$.

Orthogonalisation. $XH^\perp$ is the residual from regressing XH on log IV, log half-spread, strike count, days to expiry, BKM skew and BKM kurtosis. This strips out width, the higher moments and the measurement-noise channel from Section 4. In the full sample the regression has $R^2 = 0.616$, so 38% of the variance of XH is unique to it. Out of sample, this projection is re-estimated inside each training window to avoid look-ahead.

Evaluation. Forecasts are expanding-window from January 2018, refitted monthly with at least 200 training observations. Each target spans the next 21 trading days, so at a refit on date t the training rows dated $t - 20$ to $t - 1$ have targets that run into the test month. The last 21 training rows are therefore dropped at every refit (an embargo). Accuracy is measured by out-of-sample R^2 on log variance and by QLIKE on the variance scale (Patton 2011), evaluated at $\exp(\text{forecast} + s^2/2)$, where s^2 is the training residual variance, so that a log forecast becomes a forecast of mean variance. Nested models are compared with the Clark and West (2007) test, one-sided for improvement, with HAC standard errors (Newey and West 1987) and 21 lags. Table 2 lists the safeguards against a false result.

6 Results

Implied volatility adds 13.9 percentage points of out-of-sample R^2 to HAR; the BKM moments and excess entropy each make forecasts worse (Table 3).

Three readings follow.

- **The gate passes.** IV beats HAR with $t = +4.77$, consistent with Christensen and Prabhala (1998), who found the same for S&P 100 options. In sample, IV adds 12.3 percentage points of R^2 . The pipeline can therefore detect a real forecasting effect when one exists.
- **Shape adds nothing beyond width.** The BKM moments reduce out-of-sample R^2 by 0.015 and excess entropy by a further 0.011. Clark–West tests only for improvement, so

Table 2: Safeguards against a false result

Safeguard	Rule	Why
Calibration gate	HAR $\rightarrow$ HAR+IV must be significant out of sample before any entropy result is accepted	Reproduces a known result; failure means a pipeline defect, not evidence
Placebo	One fixed draw of pure noise through the identical machinery must be insignificant; separately, 200 simulated persistent null regressors measure the test’s actual size	Detects leakage or overfitting in the test itself
Power analysis	Plant a signal with $XH^\perp$ ’s first-order persistence, 200 replications per effect size	Turns “no effect found” into “no effect larger than X would likely have been missed”
Embargo	Drop training rows whose 21-day targets overlap the test month	Prevents look-ahead through overlapping targets

Table 3: Out-of-sample forecasts of 21-day log realised variance, January 2018 to July 2025

Model	MSE (log)	R_{oos}^2	QLIKE	Increment	Clark–West t
HAR	0.7403	0.1916	0.566		
HAR + IV	0.6129	0.3307	0.526	+0.1391	+4.77
HAR + IV + BKM	0.6265	0.3158	0.526	−0.0149	−1.07
HAR + IV + BKM + $XH^\perp$	0.6370	0.3044	0.533	−0.0114	−1.76

Source: horse_race.py, written to results.json. $n = 1,905$. Expanding window refitted monthly with a 21-row embargo; entropy extracted with Cboe strike selection. Increment is the change in R_{oos}^2 from the previous row. QLIKE at the bias-corrected mean forecast. Clark–West one-sided for the larger model, HAC with 21 lags.

Placebo $t = +0.08$.

the negative t -statistics are no evidence of predictive power in either direction. Because the statistic already corrects for the noise of estimating extra parameters, a value of -1.76 , against a simulated null mean of -0.40 , points to an unstable coefficient rather than pure estimation noise. In sample, the coefficient on $XH^\perp$ is -1.14 with a Newey–West t of -0.65 .

- **QLIKE mostly agrees.** HAR + IV has the lowest bias-corrected QLIKE (0.526); adding BKM leaves it unchanged and adding entropy raises it to 0.533. On uncorrected QLIKE, which treats the log forecast as a median, the BKM (0.657) and entropy (0.661) models score slightly better than HAR + IV (0.669), so this ranking depends on the bias correction.

Figure 1 plots implied volatility and excess entropy over the full sample.

7 Robustness

The null survives an independent re-implementation, every horizon from 5 to 63 days, the exclusion of 2020, winsorised outliers, dropping the BKM moments, the zero-return floor, and both earlier specifications (Table 4).

Independent re-implementation. verify_panel.py rebuilds the panel and the test in separately written code from the extracted entropy files and raw SPY returns; it does not re-extract entropy. It differs from horse_race.py in two ways: its orthogonalisation starts after a 250-row burn-in, so training begins in March 2016 rather than February 2015, and it uses real-time residuals rather than re-residualising the training rows at each refit. The two agree on sign and

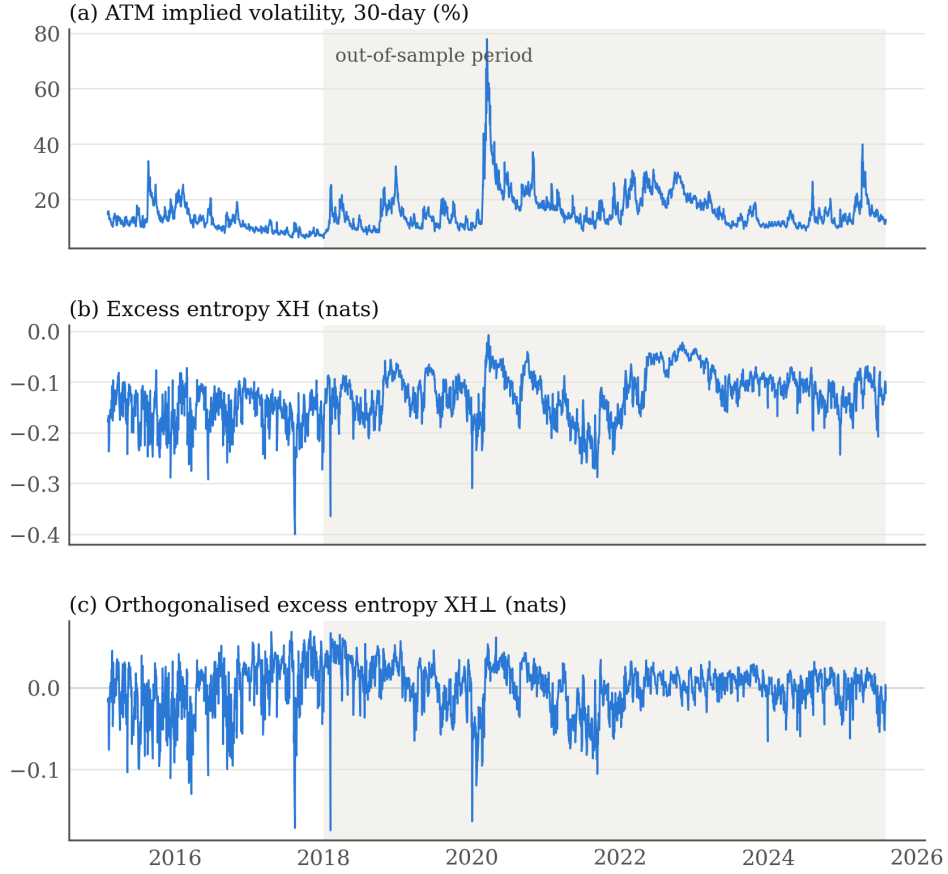


Figure 1: Implied volatility and excess entropy, February 2015 to July 2025. (a) 30-day ATM implied volatility. (b) Excess entropy XH , extracted with Cboe strike selection. (c) $XH^\perp$, the full-sample residual of XH on log IV, log half-spread, strike count, days to expiry and the BKM moments. Shading marks the out-of-sample period. XH moves with volatility regimes, rising towards zero in the 2020 and 2022 stress periods, but the part not explained by width and the BKM moments (c) is small and drifts slowly; the sharpest dips (August 2017, 5 February 2018, January 2020) survive every quote filter tested.

Table 4: Robustness of the entropy result

Check	Entropy t	IV t	Source
Main specification	-1.76	+4.77	horse_race.py
2020 excluded from evaluation ($n = 1,652$)	-1.78	+5.66	horse_race.py
XH winsorised at 1st/99th percentile (full-sample cut-offs)	-1.84		verify_bkm.py
XH and BKM winsorised	-1.65		verify_bkm.py
BKM dropped from the model and the orthogonalisation	-1.63		verify_bkm.py
Independent re-implementation	-0.59	+4.77	verify_panel.py
Re-implementation, floor at 10^{-12} / at the 1st percentile	-0.61 / -0.56	+4.77 / +4.80	verify_rv_sensitivity.py
Before the zero-bid filter (embargo, floor)	-1.63	+4.77	horse_race.py, pre-filter files
Original August specification	-0.22	+4.89	horse_race.py published, pre-filter files

insignificance but not on magnitude ($t = -0.59$ against -1.76). One candidate explanation is drift: XH ’s annual mean across all date-expiry pairs moves from -0.19 in 2015 to -0.09 in 2022, so a coefficient fitted on early data may misfire later, and the start of training would then matter. This was not tested directly.

Horizons. The null holds at every horizon tested, with the embargo set to h rows and HAC lags to $\max(h, 21)$. IV’s advantage decays with horizon, as it should (Table 5).

Table 5: Clark–West t -statistics by forecast horizon

Horizon (trading days)	5	10	21	42	63
Entropy t	−0.46	−0.64	−0.59	−0.20	−0.25
IV t	+7.08	+6.20	+4.77	+3.76	+3.14

Source: `verify_panel.py`.

Power. $XH^\perp$ is persistent ($\text{AR}(1) = 0.82$) and the 21-day targets overlap, so the sample carries far fewer independent observations than its row count. Applied to the 2,369 rows from March 2016, an $\text{AR}(1)$ approximation gives about 240 effective observations; this is optimistic, because the autocorrelation decays slowly (0.35 at 21 days). The Monte Carlo (`verify_power.py`) plants a signal with $XH^\perp$ ’s first-order persistence into the real panel, explaining a share k of the residual variance after HAR + IV + BKM, and runs the re-implemented embargoed out-of-sample test 200 times per value of k (Table 6).

Table 6: Power of the out-of-sample test against a planted persistent signal

Planted share k	Power ($\pm$ s.e.)	Out-of-sample incremental R^2 achieved
0 (null)	0.060 ± 0.017	−0.0039
0.015	0.55 ± 0.04	+0.0070
0.025	0.65 ± 0.03	+0.0108
0.030	0.77 ± 0.03	+0.0163
0.040	0.88 ± 0.02	+0.0214

Source: `verify_power.py`.

The test rejects 6.0% of the time under the null at a nominal 5%. Power reaches 80% between $k = 0.03$ and 0.04; interpolating, a signal worth about 0.018 of out-of-sample R^2 , roughly one-eighth of IV’s contribution, is detected with 80% probability. The detectable effect is larger than before the zero-bid fix (about 0.01): quote noise had made $XH^\perp$ look less persistent ($\text{AR}(1)$ 0.58 against 0.82), which overstated the effective sample and so the test’s power. The simulation uses the re-implemented test; the headline test differs mainly in its training start, and its power was not simulated separately.

Numerical stability. Runs on macOS print matrix-multiply warnings from Apple’s Accelerate library. In August, `check_orth.py` found full column rank and finite coefficients in the out-of-sample orthogonalisation for the 2018–2019 windows, and the August results reproduce exactly to four decimals, so the warnings do not affect the arithmetic.

8 Pipeline validation: eight defects

Eight defects were found and fixed (Table 7). The calibration gate exposed the first five during the project, before any entropy result was accepted. A fresh review of the repository before

write-up found the last three. Each traced to data or code, not to the hypothesis. The figures for defects 1 to 5 come from August runs recorded in the research log; the repository documents the fixes but does not reproduce those diagnostics.

Table 7: Defects found and fixed

#	Defect	Symptom	Effect if missed
1	Unsorted moneyness passed to linear interpolation	ATM IV wrong on about 1% of dates	Noisy IV understates the benchmark, flattering any rival
2	Dependent variable built from a per-unit-time microstructure variance field (ivol.t)	Needed a 15,250× rescale; corr(log VIX, forward RV) only 0.43; gate failed	Target barely related to market volatility
3	Open and close “price” columns were constant 1.0 flags	The Garman–Klass estimator (Garman and Klass 1980) silently reduced to half the squared range	Wrong variance estimator, no error raised
4	Column search matched only names containing “price”	Real price columns (OPrc, CPrc) never found	Calibration ran on the wrong fields
5	Intraday high and low contain erroneous prints	Median daily range $2.77\times$ the absolute close-to-close return, against about 2.1 for a simulated Brownian path; corrupted days dominated 21-day means	Range-based RV biased upward by a minority of bad days
6	No embargo: training rows whose 21-day targets overlapped the test month	On the pre-filter extraction, entropy t moved from -0.22 to -1.63 once removed; the change grows with the embargo and levels off at the 21-row overlap (-1.59 at 21 rows, -1.60 at 22)	Look-ahead flattering every model’s out-of-sample fit, with the entropy result’s change concentrated in 2020
7	In-sample statistics reported as out-of-sample (entropy $+0.0008$, IV 0.131 , $n = 2,369$ counted from March 2016)	Each reproduced exactly from full-sample regressions	Headline figures a reader could not recover from the out-of-sample test
8	Zero-bid quotes kept in the deep option wings, entering at half the ask	XH spikes to -0.95 and BKM kurtosis above 500; on re-extraction with Cboe strike selection every spike disappears, and on 217 sampled dates XH before and after correlates only 0.68	Noise in the entropy measure that understated its persistence and so overstated the test’s power

The fifth defect was confirmed by a discriminating test: the 21-day mean of range-based RV correlated 0.362 with VIX, while the 21-day median correlated 0.617. A few extreme days were driving the mean. This is why close-to-close returns are the variable of record, and the RV builder now asserts the 0.60 correlation gate before writing its output.

The sixth defect was confirmed by a control, run on the pre-filter extraction. Dropping 21 training rows from the start of each window, or 21 rows at random, leaves the entropy t between -0.07 and -0.35 , close to the original -0.22 . Only removing the 21 rows whose targets overlap the test period moves it. The move came almost entirely from 2020: without the embargo, excluding 2020 already gives $t = -1.31$, while with the embargo the result is the same with or without 2020 (-1.59 and -1.61).

The eighth defect was found with the help of an external benchmark. Cboe’s SKEW index (Cboe n.d.) is $100 - 10$ times the 30-day risk-neutral skewness of SPX returns, the quantity the BKM skewness estimates. After the fix, pipeline skewness correlates 0.85 with it in levels and 0.62 in daily changes, and the full-sample medians are -3.51 and -3.28 . Annual medians differ by 0.1 or less, one SKEW index point, in 7 of 11 years, and by 0.16 in 2020. In the calm years 2017 to 2019 pipeline skewness is 0.6 to 1.6 more negative. Cboe interpolates to exactly 30 days

across all SPX expiries, whereas this paper uses the AM-settled monthly expiry nearest 30 days; whether that explains the gap was not tested. The large kurtosis is consistent with these prices: kurtosis must exceed skewness squared plus one, so Cboe’s own skewness of about -4.8 in 2024 implies a kurtosis above 24, and the pipeline’s full-sample median kurtosis is 34 (Table 1).

The lesson is general. Every defect produced a number that looked plausible in isolation. Requiring a known result first exposed the first five; the last three needed a fresh read of the code, a control designed to tell leakage from data loss, and an external benchmark for an intermediate quantity.

9 Discussion and limitations

The most likely explanation is that in the S&P 500, the option market’s pricing of shape is mostly spanned by its pricing of level, and the remainder reflects the price of risk rather than its quantity.

Why shape adds nothing. Three mechanisms are consistent with the result, and they are not mutually exclusive.

- **Shape moves with level.** Index skew steepens as volatility rises, so much of what entropy measures is already in ATM IV. The orthogonalisation regression explains 62% of the variance of XH .
- **Risk-neutral is not physical.** The RND embeds risk premia. If the left tail is fat because investors pay for crash insurance (the skewness-preference channel), the shape measures demand for protection, which need not predict realised variance. A forecast of physical variance can only use the part of shape that reflects expected outcomes.
- **Measurement noise attenuates.** The entropy estimate carries quote-noise error, which biases any true coefficient toward zero. The orthogonalisation removes the noise channel that correlates with the spread, but not classical error. The zero-bid filter removed the largest identified source of such error.

Limitations.

- **One market, one maturity, one target.** The test covers SPX at about 30 days, forecasting realised variance. Shape could still forecast tail events, returns, or single-stock volatility, where skew is less tied to level.
- **Bimodality is invisible.** The harness shows two-peak densities cannot be detected at SPX quote precision, so event-driven shapes (elections, FOMC) are outside the test.
- **Sample.** Seven and a half out-of-sample years include one extreme episode (March 2020). The power analysis bounds average effects, not effects confined to rare regimes.
- **Economic significance.** The planned test of whether any forecast improvement survives transaction costs is moot given no statistical improvement.

Estimation stability. The two implementations agree on the null but differ in magnitude ($t = -1.76$ and -0.59), and the negative headline statistic points to an unstable coefficient. Drift in XH across years is a plausible cause that was not tested directly; a rolling rather than expanding window would be the natural check.

10 Conclusion

Excess entropy, a measure of the shape of the option-implied distribution of S&P 500 returns, does not improve forecasts of future volatility once implied volatility is known. The test would detect an effect worth about 0.018 of out-of-sample R^2 80% of the time, and finds none: the estimate is -0.011 . Skewness and kurtosis add nothing either. For forecasting how volatile the market will be, the width of the distribution is enough. One explanation is that the shape mostly reflects what investors pay to insure against crashes, or moves together with the width, rather than telling us anything new about the risk that actually arrives.

The main lesson for me is about method. Requiring the pipeline to reproduce a known result before trusting the new one caught five errors, and a fresh review caught three more; without those checks I would have reported a wrong number with confidence. A null result is only informative when the test is shown to work and its power is stated, and this paper tries to do both. Natural next steps are to test shape against crash risk or returns rather than volatility, on individual stocks, and on short-dated options around scheduled events such as central bank meetings.

Code and the verification suite:

<https://github.com/samlessan/spx-entropy-volatility>.

Acknowledgements and use of AI tools

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Use of AI tools. I used an AI assistant, Claude (Anthropic), as my coding and drafting assistant throughout the project. Claude wrote most of the Python code, including the implementation of the checks I proposed; proposed much of the technical design, including the orthogonalisation and the model-free moment controls; diagnosed and fixed the defects described in Section 8; and produced the first draft of this paper. My own contribution was as follows. I developed the research question and the entropy measure in discussions with my supervisor, who first proposed using entropy. I secured access to the licensed data through Warwick Business School and downloaded the option and intraday data myself. I proposed the calibration gate and the placebo control. Because the data are licensed, I ran every script on my own machine, so every reported number comes from program output I produced, and I reviewed each output and asked for changes where it fell short. I spotted the first erroneous price print in the SPY data, which led to defect 5. I decided which methods and corrections to adopt, and I chose to report the null result rather than search for a specification that produced a significant one. Before submission I had Claude review the whole project, and required each suspected error to be confirmed before anything was changed. I then manually revised the draft. Defects 6 to 8 were corrected after the main results had been seen, so the paper (Table 4) and the repository’s corrections log report the result under each specification. The conclusions are my own.

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